

Team Number 10

Project Title: Intelligent and Collaborative Autonomous Robots for Real-Time Applications

Report Date: 9/9/2024

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.
 - a. Yes, team was able to meet in lab and over teams
 - i. John and Ian teams meeting on 9/8
 - ii. All members present in lab 9/10 at 5pm
2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.
 - a. Yes we have met with the team sponsor.
 - i. All members in attendance in the lab 9/10 at 5:30 PM
3. Describe verbally the tasks completed the past week and the challenges faced.
 - a. This week our tasks were to put the YOLO model onto a local computer this will help with run times as we are not relying on a limited amount of GPU. It seems as though Yolo only recognizes NVIDIA graphics cards. One of the challenges faced was trying to run the Yolo model on an AMD GPU which was unsuccessful.
 - b. We were successful in being able to incorporate our YOLOmodel over a live video feed from the webcam on the computer which came with decently accurate results in identification
 - c. We were also able to explore the RaspberryPi on the robot and understand where to find processes attached to the robot as well as access the directories within the storage.
4. Describe the tasks to be completed the coming week.
 - a. The robot needs network access in order to download the repositories the goal is to explore the files within the storage for steps that may have been taken to grant network access for downloading.
 - b. We will be attempting to locate the source of the camera within the dog for us to attach our code to see a live view of the camera directly from the Raspberry Pi
 - c. on the side we will also be testing the command ability of the robot over code for decision making as well

Copy the following table from the last week's report (09/02)

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
David Uzan	2	0	2
Ian Simmons	2	0	2
John Sinfuego			

Fill in the following table for this week: (09/09)

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
David Uzan	2	0	2
Ian Simmons	2	0	2
John Sinfuego			

Part 2: Individual Progress Report (each member complete one form)

Name of the Member: Ian Simmons

Report Date: 9/9/2024

List the following:

1. All tasks completed the week before the past week (completion date in parenthesis). This is copied from the last report, and if this report is the first one, skip.
2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **before the past week**:

1. **Week 2, Task 1 (9/2)**
 - a. Research different nanos on Go1 as well as RaspberryPi
 - i. This task consisted of understanding what we were using to run our tests as well as filtering out any unnecessary information for this project
 - ii. It is determined that we are using nano1 for the front camera
2. **Week 2, Task 2 (9/2)**
 - a. Research a way to run YOLO on other brand of GPU for personal testing
 - i. Primarily Yolo incorporates NVIDIA graphics cards as their form of processing not everyone has this available
 - ii. For this specific project using code that uses recognizes other graphics cards, specifically AMD was found to not be suitable for this project,

For **each** task you have completed **during the past week**:

3. **Week 3, Task 1 (9/9)**
 - a. Research ways of adding Wi-Fi to robot
 - i. One of the key issues is the inability for the dog itself to maintain access to the network wirelessly. There is no current direct approach however there does exist an adapter for this process. One option is to access the Pi's config directly

```

Raspberry Pi Software Configuration Tool (raspi-config)

1 Change User Password Change password for the current user
2 Network Options       Configure network settings
3 Boot Options          Configure options for start-up
4 Localisation Options  Set up language and regional settings to match your
5 Interfacing Options   Configure connections to peripherals
6 Overclock             Configure overclocking for your Pi
7 Advanced Options      Configure advanced settings
8 Update               Update this tool to the latest version
9 About raspi-config    Information about this configuration tool

<Select>                                <Finish>

```

```

network={
    ssid="YOUR_SSID"
    psk="YOUR_PASSWORD"
    id_str="SCHOOL_NETWORK"
}

```

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
 ^X Exit ^R Read File ^\ Replace ^U Uncut Text ^T To Spell ^_ Go To Line

4. Week 3, Task 2 (9/9)

- a. Connect to RaspberryPi and explore directories
 - i. The goal of this is to familiarize ourselves with the procedure for navigating through the libraries stored on the robot whilst practicing Linux commands at the same time. This is the main way to navigate the robots on board system to locate points of interest as well as introducing our own software/code

```

pi@raspberrypi:~ $ ls
MagPi  PythonTest
pi@raspberrypi:~ $ cd ..
pi@raspberrypi:/home $ cd ..
pi@raspberrypi:/ $ ls
bin  dev  home  lost+found  mnt  proc  run  srv  tmp  var
boot  etc  lib  media  opt  root /sbin  sys  usr
pi@raspberrypi:/ $ cd etc
pi@raspberrypi:/etc $ cd wpa_supplicant/
pi@raspberrypi:/etc/wpa_supplicant $ sudo nano wpa_supplicant.conf

```

ii.

For **each** task you are currently working on or plan to work on in the next period:

1. **Week 4, Task 1 (9/16)**
 - a. Explore files within robot for usability in network access
 - i. It has been noted that there is a way to give access to the robot and it is possible that it is located within one of the directories within storage.
 - ii. If found this may be a great start to setting up networking capability
2. **Week 4, Task 2 (9/16)**
 - a. Find/Register Mac address to local network
 - i. This is another method that is an option for giving network access to the robot
 - ii. The significance of having network capability is for us to be able to download repositories directly to the robot as the program is running. This will greatly speed up the time needed for computing operations

Team Number 10

Project Title: Intelligent and Collaborative Autonomous Robots for Real-Time Applications

Part 2: Individual Progress Report (each member complete one form)

Name of the Member: David Uzan

Report Date: 9/9/24

List the following:

1. All tasks completed the week before the past week (completion date in parenthesis). This is copied from the last report, and if this report is the first one, skip.
2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **during the before the past week**:

1. Task number/name

General Task 2 (9-6-24) – Implement code for video recording on dog and use it to collect video data from the Uni-Go1.

Personal Week 2 (9-6-24) – Using video recorded by the Uni-Go1, I will apply the detection algorithms to it to find exactly what the robot detects with its cameras.

For **each** task you have completed **during the past week**:

1. Task number/name

General Task 3 (9-13-24) – Establish baseline for use cases and testing and search for methods to run YOLO directly on Uni-Go1

Personal Week 3 (9-13-24) – Migrate object detection from Google Colab to local IDE and test live feedback capabilities of the Uni-Go1

2. Short description – these descriptions form the basis of the final project report
 - a. Implementation, testing, results

Using documentation from the Ultralytics GitHub and home page, I set up YOLO to run on Visual Studio Code first testing with prerecorded video and then changing the input source to be my personal webcam for real time object detection. The results were taxing for my computer as it is not meant to handle harsh computing like the algorithm used, but the results were shown in the screenshots below, restricted to display objects that are detected above 60%confidence.

b. Brief and to point

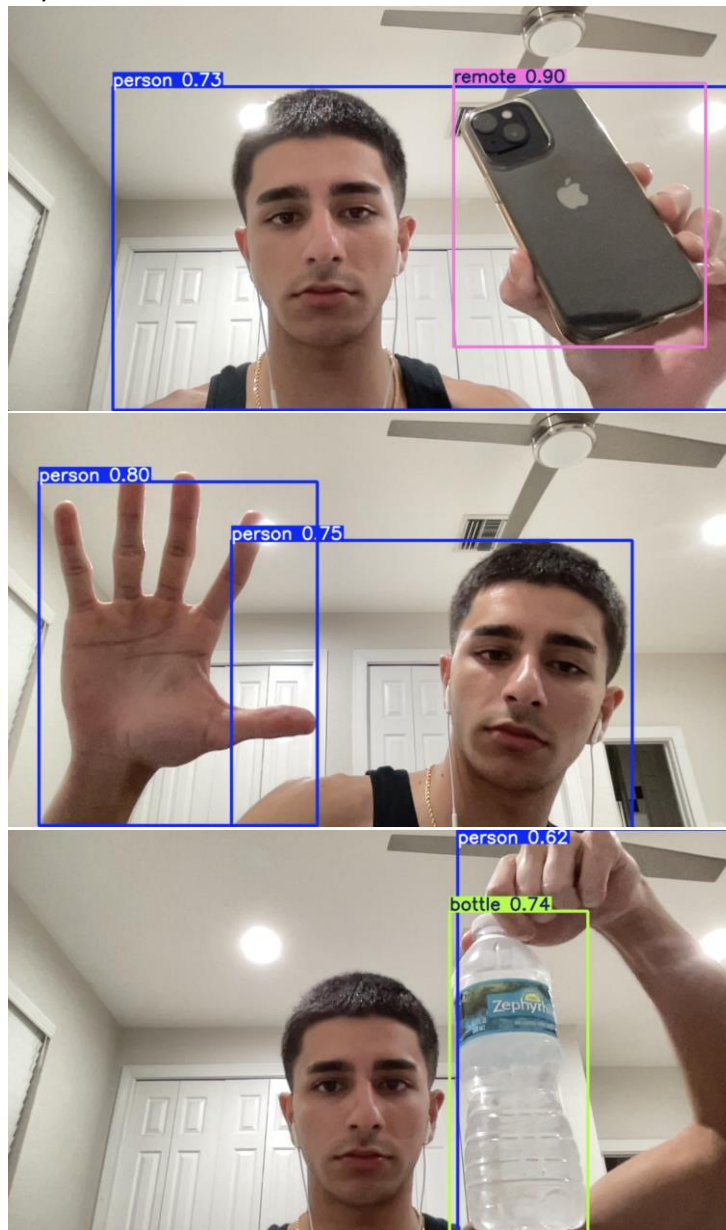
I wrote new code to run on Visual Studio Code instead of Google Colab and ran it using my webcam for quick results in real time.

3. Outcome of the task activity

a. How did the task advance your project?

The task advanced the project greatly because we now know that the YOLO algorithms can be used in real time situations, although they yield varying confidence and detection.

b. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task



For **each** task you are currently working on or plan to work on in the next period:

1. Task number/name

General Task 4 (9-16-24) – Incorporate the model into Uni-Go1 and run tests across multiple cameras.

Personal Week 4 (9-16-24) – Locate camera source directory in Uni-Go1 and display live video feed from it.

2. Short description

a. Plans, resources needed, implementation, results

Using the manual of the Uni-Go1 and previous projects' work to access storage locations inside of the robot are crucial for the next step of implementation. Once the locations are found and linked to their function, we will begin installation of necessary software and run basic tests to get a more definitive answer of how the robot will detect objects in its environment.

b. Main challenges, reasons for any delay in completion

A previous project using the same Uni-Go1 model found that accessing the cameras of the robot killed the process until it was reset. We would have to spend more time figuring out a way that does not break the camera feed from the robot to increase reliability of the system.

3. Outcome of the task activity

a. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task

N/A, have not started currently.

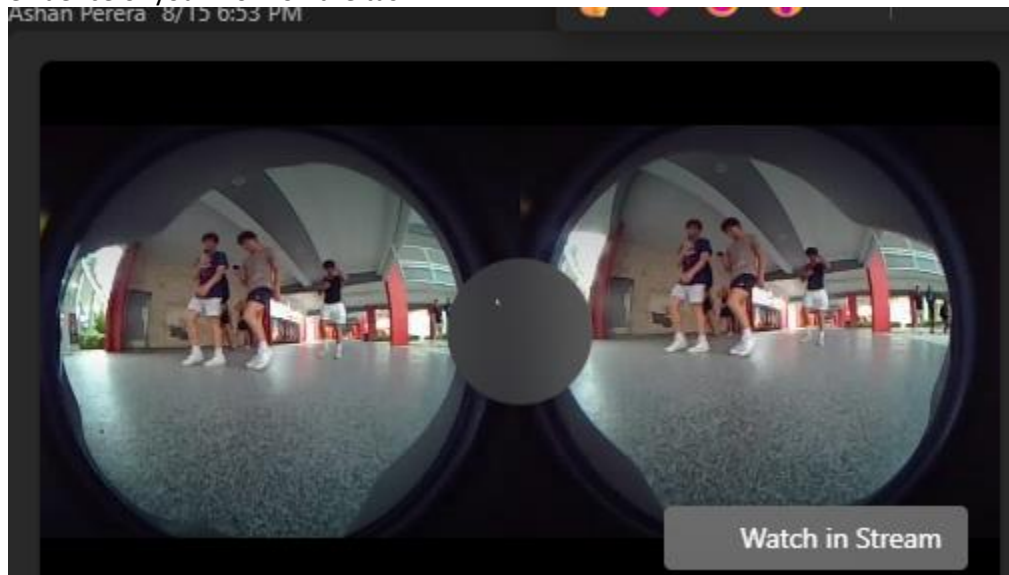
Name of the Member: John Angelo Sinfuego

Report Date: 9/9/24

List the following:

For **each** task you have completed **during the past week**:

2. Short description – these descriptions form the basis of the final project report
 - a. I met up with my team members in campus to learn how to use and get familiar with Go1 as well as record a video on campus. We also learned how to use the YOLO model and record a video in order to establish a baseline.
 - b. Brief and to point
We learned more about go1 capabilities as well as learned more on the code that can transfer the image from go1 to a computer.
3. Outcome of the task activity
 - a. How did the task advance your project?
This advances our project a significant amount because now we can start implementing object detection with our videos we recorded with go1.
 - b. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task



For **each** task you are currently working on or plan to work on in the next period:

1. Task number/name

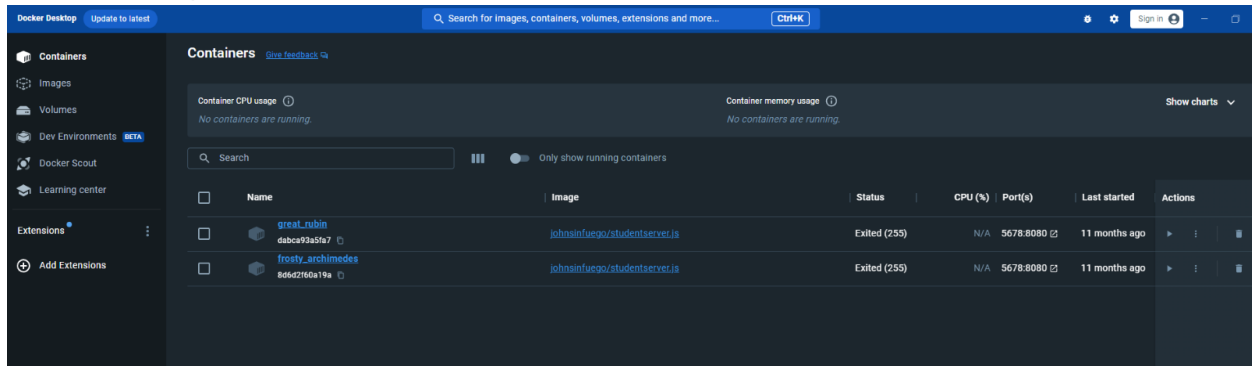
General Task 3 9-9-24 - Establish Baseline for use cases and testing

Personal Week 3 9-9-24 - Search for methods to run YOLOv8 directly from go1

2. Short description

- a. We have to run the YOLOv8 model directly from the unigo1 robot in order to establish object detection with our live feed. Since this, we have been researching different methods in order to figure out how to accomplish this task.
 - b. There is no wifi on the unigo1 robot, which makes it challenging to download the YOLOv8 repository into it because this needs wifi in order to be done. I have researched more methods to accomplish this, via different wifi adapters to put in with raspberry pi, and using a program called docker in order to establish code into the raspberry pi so it can be transferred into go1.
3. Outcome of the task activity

Docker desktop-



For **each** task you are currently working on or plan to work on in the next period:

1. Task number/name

General Task 4 9-16-24 - Incorporate our model onto unitree, Run tests in order to access camera

Personal Week 4 9-16-24 - Search for methods to run YOLOv8 directly from go1

2. Short description

a. We must incorporate our YOLOv8 model into unitree so we can access the yolo repository into our robot's live footage as well as run tests so we can access our camera. I have researched and found some solutions with using docker being one of the main solutions. Also put in research into other Wi-Fi adapters that may connect to raspberry pi as well as looking into a previous team's experience on working on Wi-Fi on the dog. This is found within the files in unigo1.

b. For our challenges, we still have trouble incorporating our YOLOv8 repository into unigo1. Since our dog does not have wi-fi, we still have to find a solution for this and are still researching it. We figured out that

3. Outcome of the task activity
N/A